

Heritage Digital Twin Ontology in action

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**UK Research
and Innovation**

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Use Case: The Derynia icon



The Derynia icon

The icon originates from the church of the Birth of the Virgin, built in the 15th / 16th century in Deryneia, Cyprus. The white frame delineates the XRF scanning area.

Art Characterisation

Biography

Genre
History
Visual analysis
Intangible aspects
Related artworks

Provenance

Ownership
Bibliography
Access
Attestations

Conservation

Treatment
Restoration
Prevention

Physical properties

Structure
Dimensions
Materiality
Technique
Preservation



Technical metadata of the in-house built MA-XRF, RIS, LIS scanner

- Experimental conditions
 - Data acquisition: 365-nm;
 - Size area: 25 x 17 cm²;
 - Step size: 500- μ m step size / 80-ms of dwell time per pixel;
 - Time of scan: 3h30 min;
 - XRF - X-Ray generator operated at 40 kV and 500 μ A
 - RIS - reflectance spectroscopy range 470-820 nm (UV-Vis-NIR)
 - LIS - luminescence excitation wavelength - 365 nm, 655 nm
- Raw data:
 - Raw datacube size: 498 pixels (h) by 340 pixels (w);
 - Total channels 2600: CXRF=512 CRIF=1044 and CLIS=1044
- Processing software: PyMCA
- RIS:
 - Python library pysptools 0.15.0 and SPECTRONON software.
 - Endmembers extracted using NFINDER algorithm implemented in the pysptools library.
 - Spectral Angle Mapper (SAM) was performed with SPECTRONON software.
- LIS: LIS datacube processed with PyMCA software

Lead white

Observations

Low energy Pb-M lines (2.3 keV) relate to surface lead emission. High energy Pb-L lines (10.55 keV) relate to lead deeper within the paint layers. The PyMCA RGB correlator qualitatively assess Pb-L and Pb-M intensity lines: blue denotes a deeper presence of lead (Pb-L signals, not Pb-M), magenta the presence of Pb-L and Pb-M lines, *i.e.* surface lead.

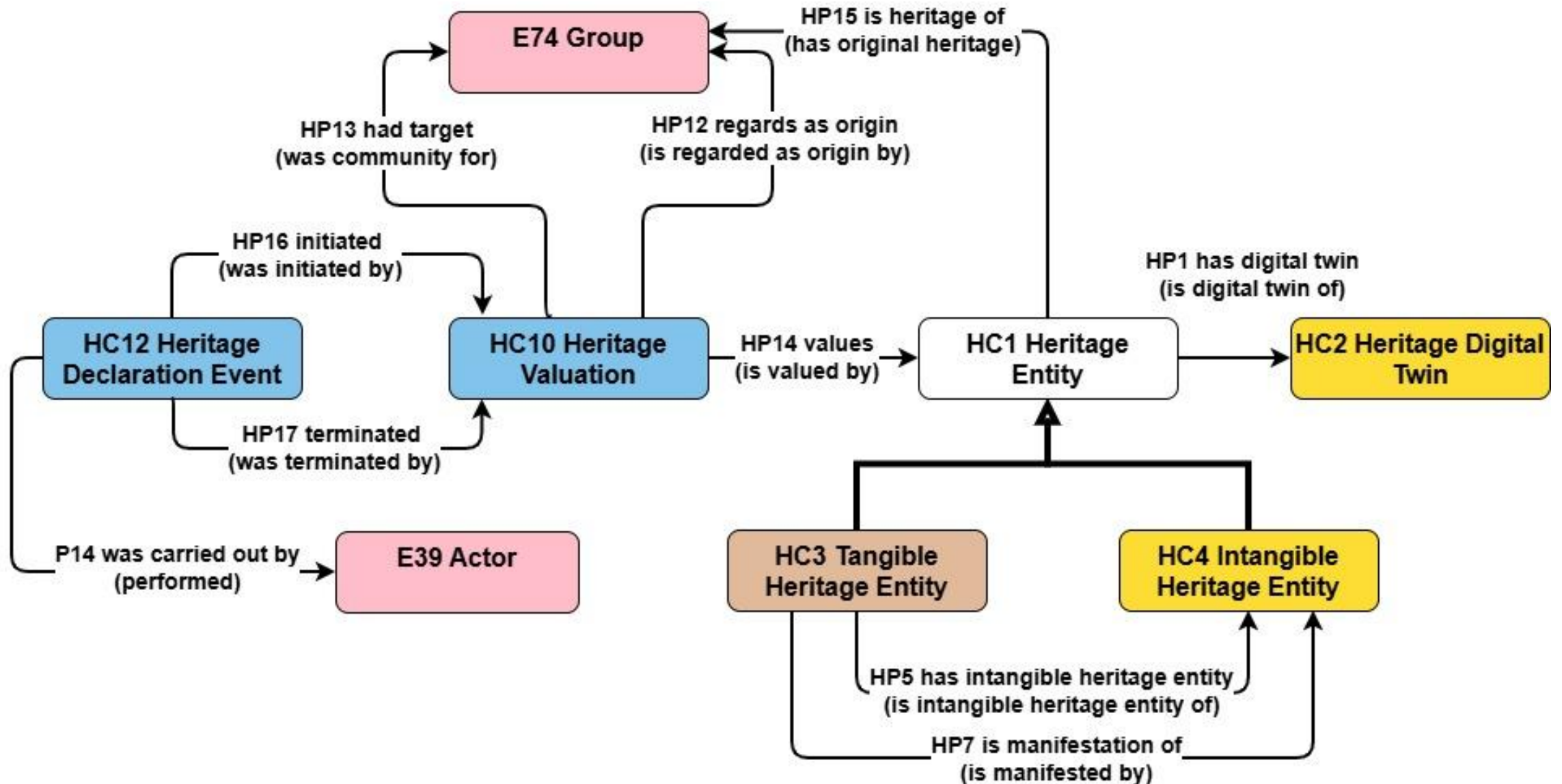
Derivate Statements

Lead white was used to tint flesh and clothing folds. The cave, the star rays, Mary's mattress edges, have several layers of lead white (blue areas), magenta - a single surface layer.

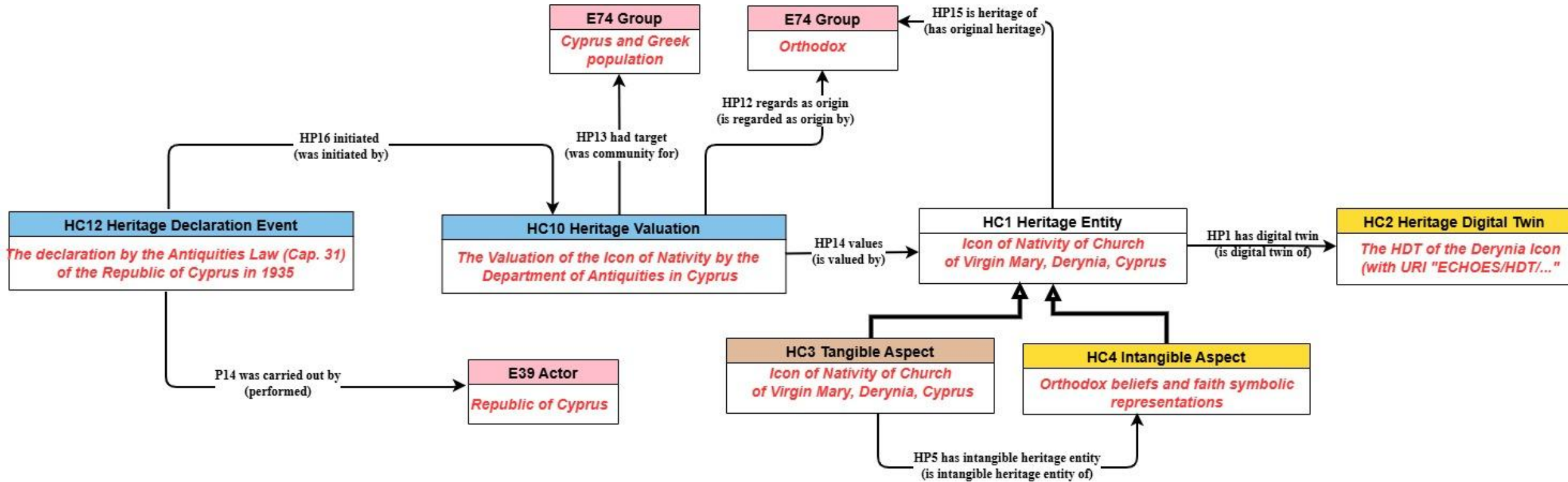


Heritage Valuation

Heritage Valuation



Derynia - Heritage Valuation



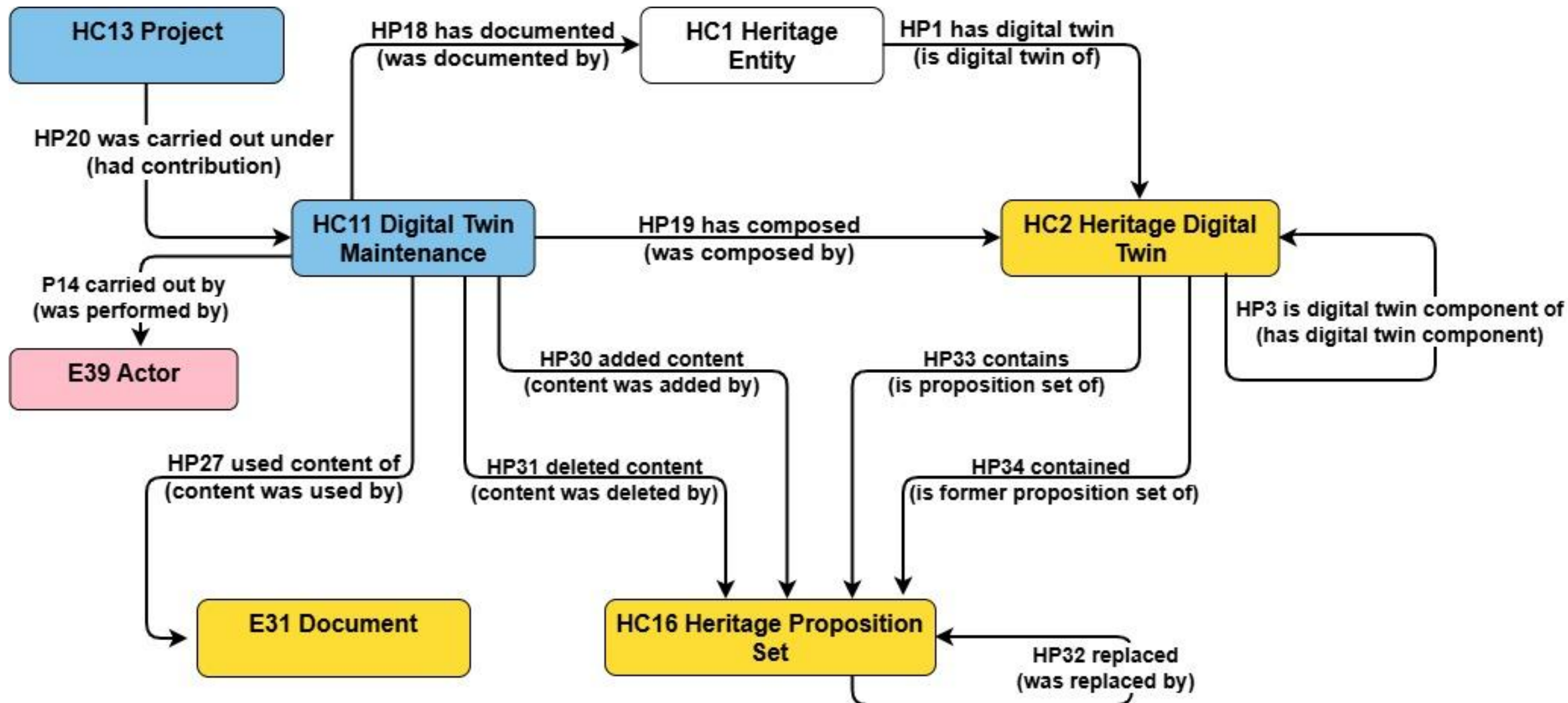
Derynia - Valuation

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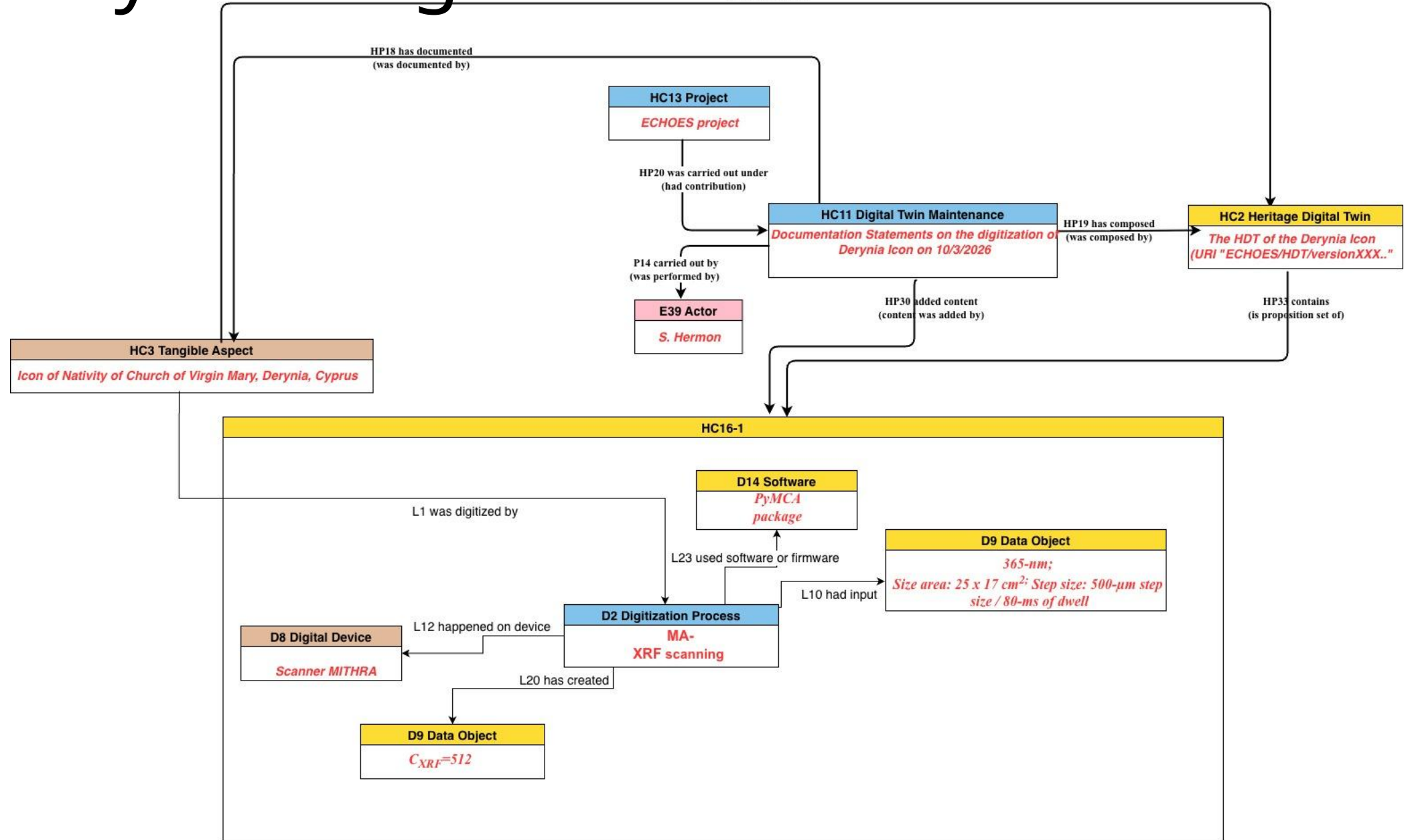
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Digital Twin Maintenance

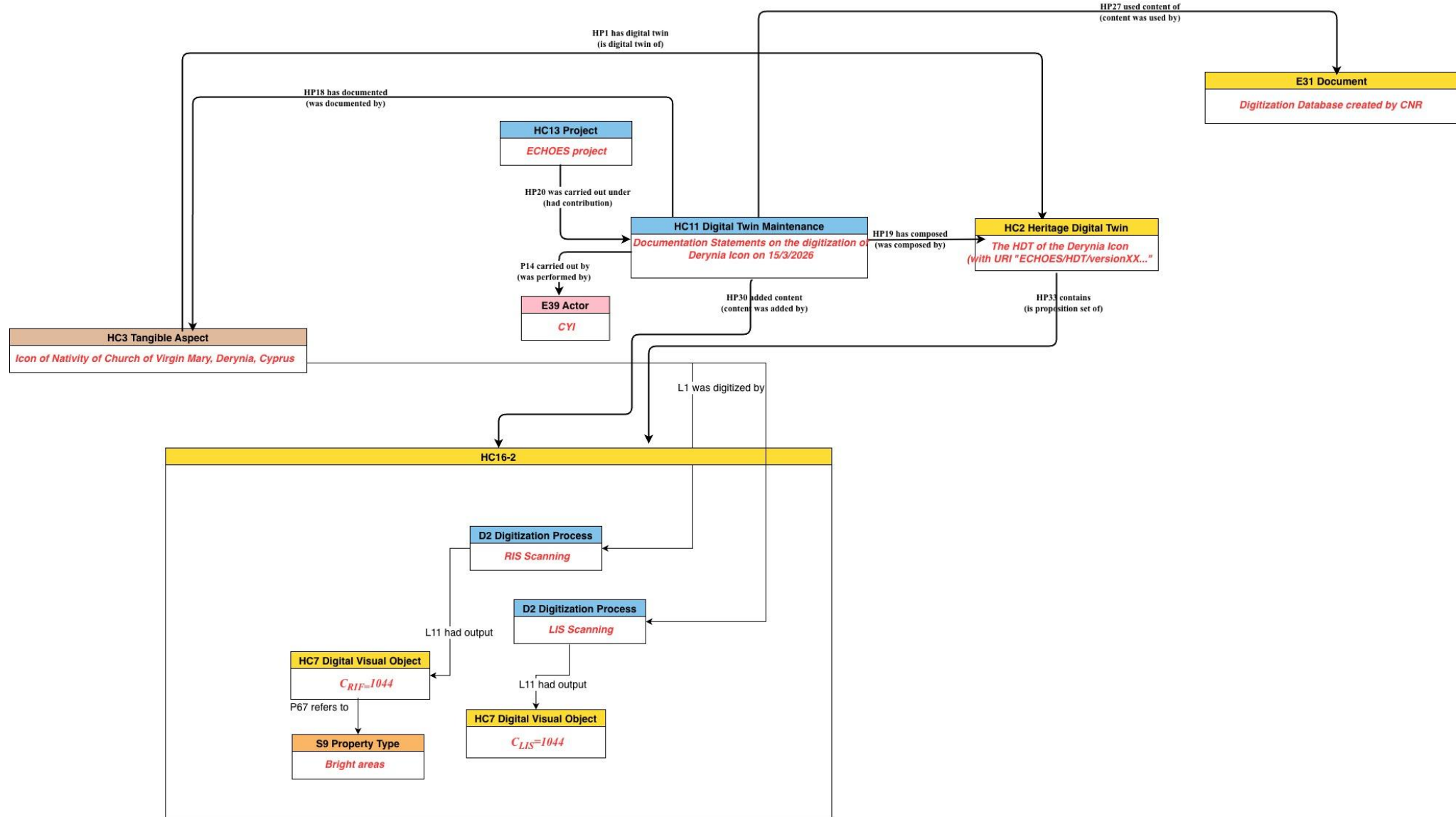
Digital Twin Maintenance



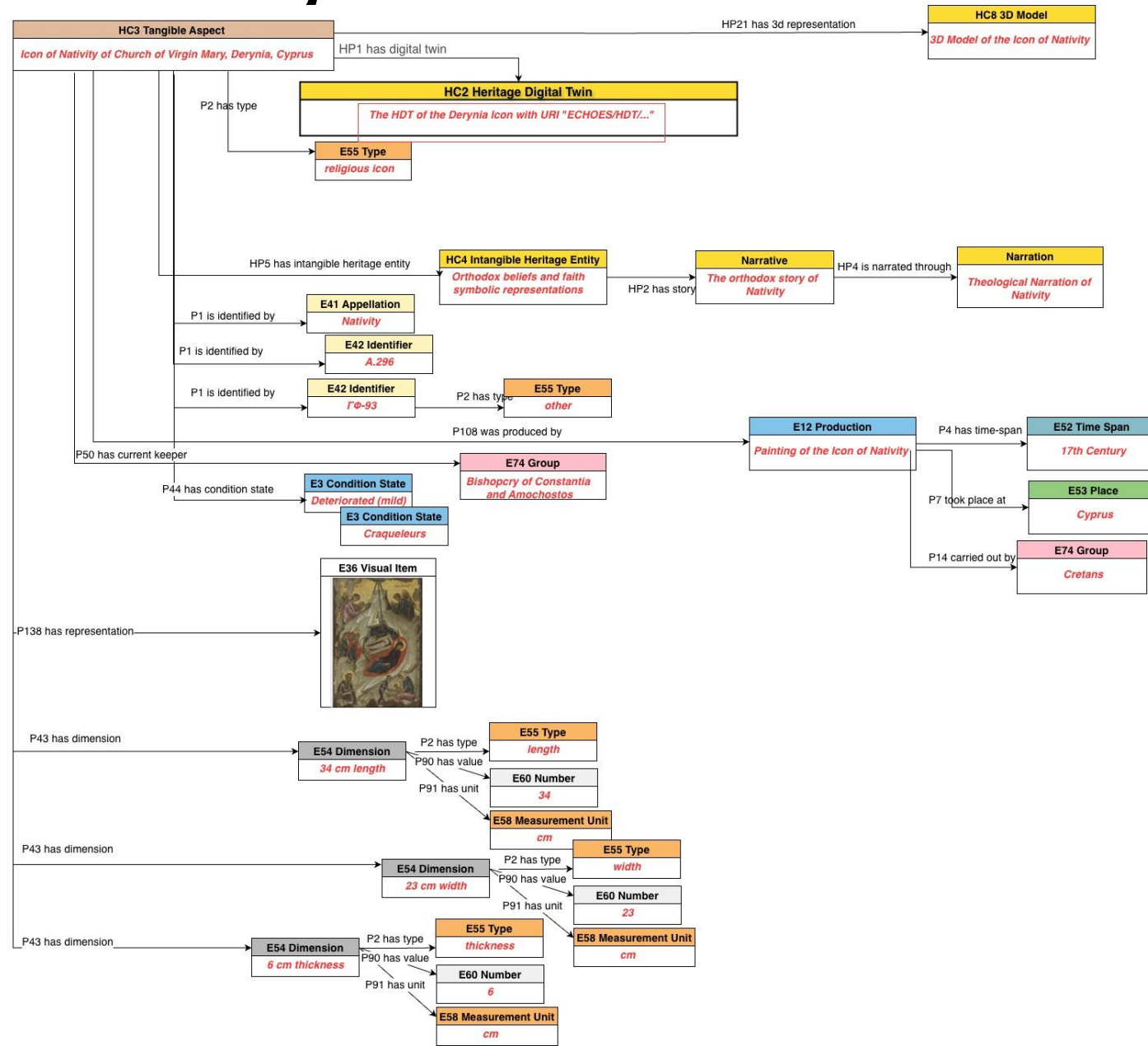
Derynia - Digital Twin Maintenance 1



Derynia - Digital Twin Maintenance 2

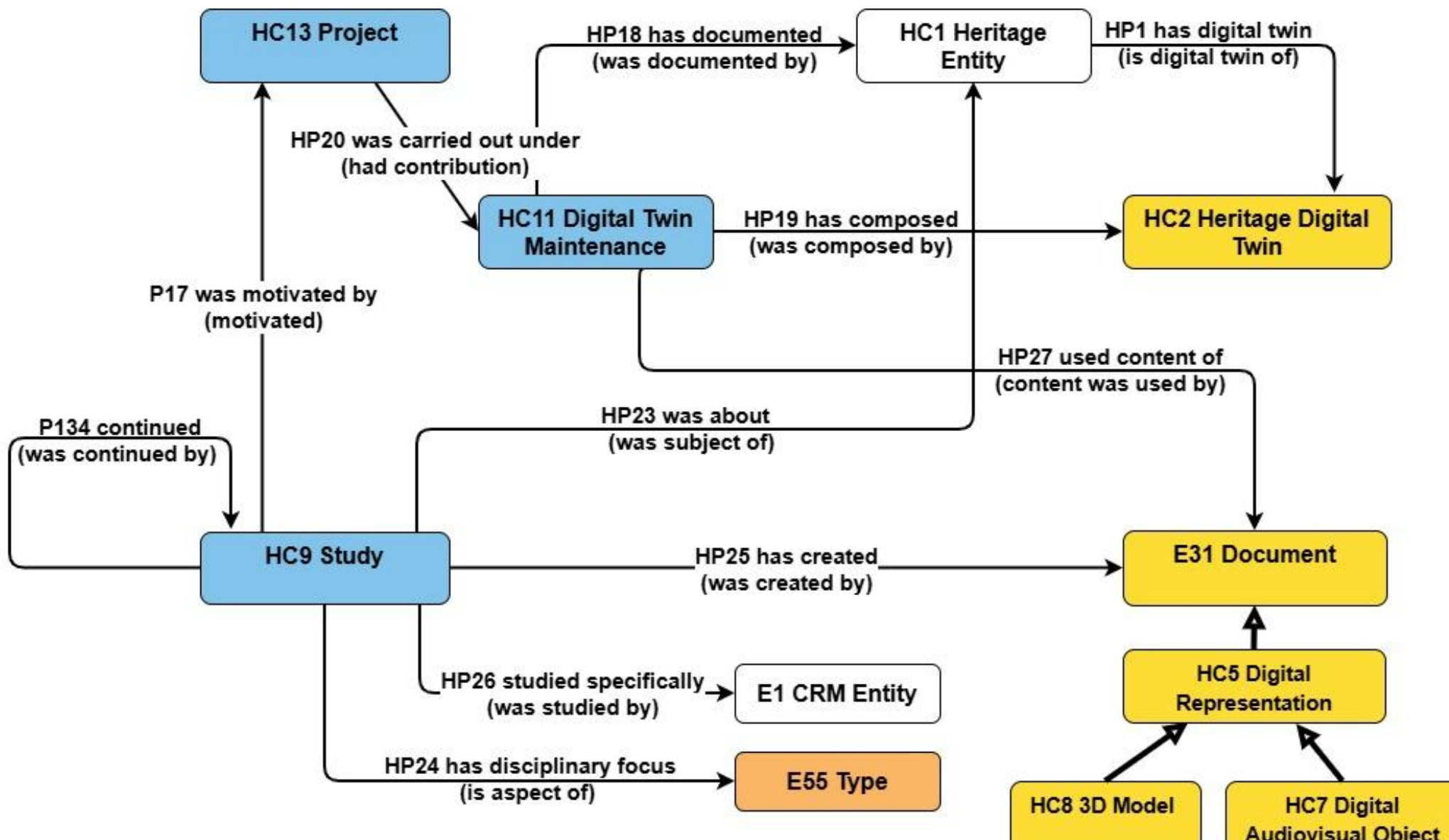


Derynia - Digital Twin Maintenance 3



Study

Study



Thank you for your attention!

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 Questions? 



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